

Final Report - Outline

Project III - Tools for Program Development

1. Introduction

Present the project context, problem, and solution.

2. Objectives

- Build a working web-based task and bug tracker.
- Use modern development tools.
- Demonstrate traceability through Git.
- Provide documentation and a functional demonstration.

3. Technologies Used

Explain React, Vite, Node.js, Express, SQLite/sql.js, Git, and Mermaid.

4. Requirements

Include the functional and non-functional requirements from Report 1.

5. System Architecture

Describe the client-server architecture and include diagrams.

6. Database Design

Describe users, projects, issues, comments, and activity tables.

7. Implementation

Explain the main modules:

- Authentication
- Project management
- Issue management
- Comments
- Dashboard
- Filtering and search

8. Testing

Include:

- Manual testing checklist
- Backend security test
- Frontend build verification

9. Functional Demonstration

Use `docs/demo-script.md` as the demo flow.

10. Conclusions

Summarize the achieved functionality and possible improvements.

11. Future Improvements

- Real user registration
- Password reset
- File attachments
- Notifications
- GitHub issue synchronization
- Advanced reports